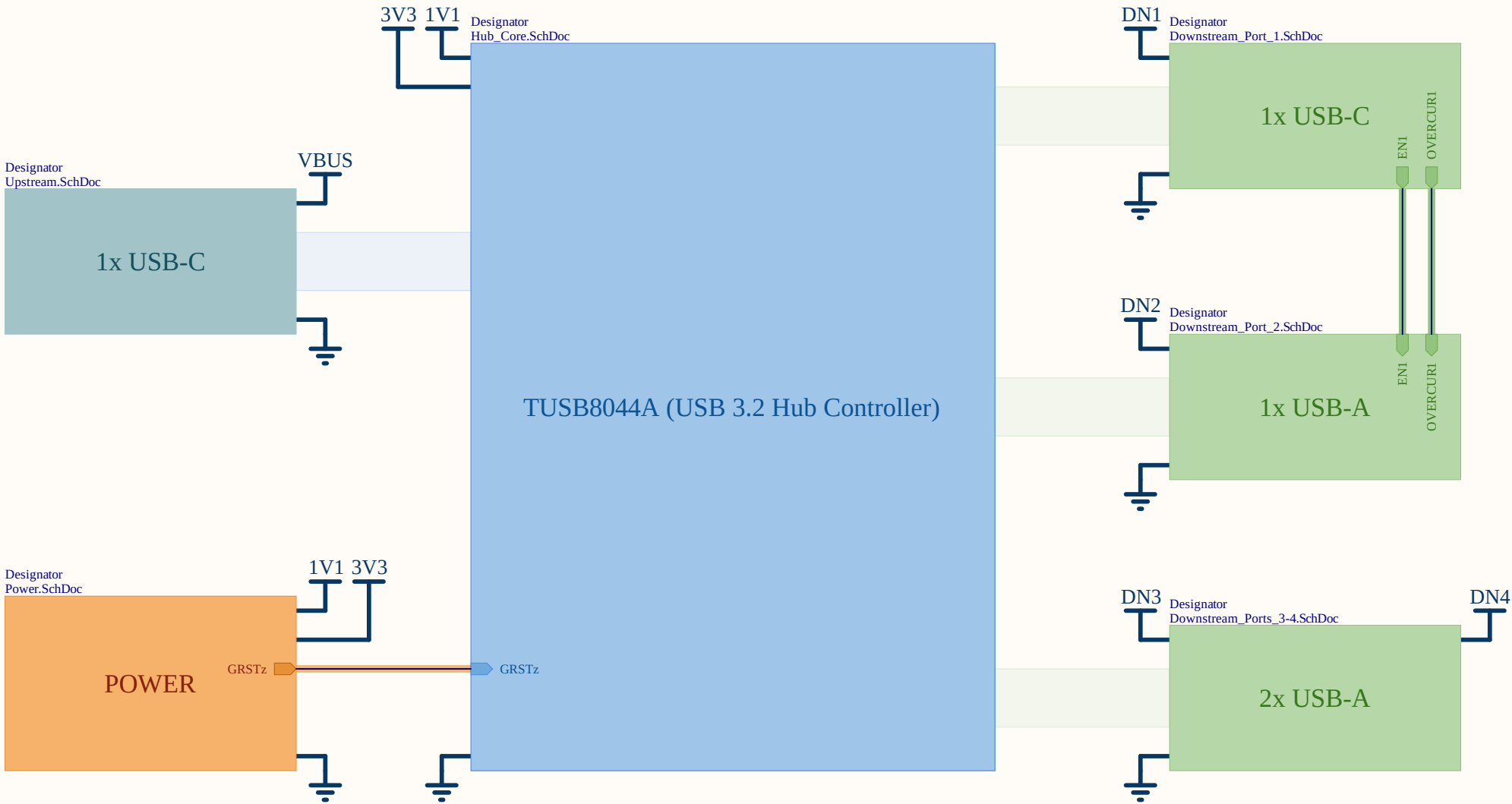


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USB 3.2 Gen 1 4-Port Hub



POWER BUDGET - USB 3.2 Gen1 Hub Bus-Powered, No USB PD		
INPUT		
USB-C Upstream (5V, 3A max)	15.00 W	3000 mA
HUB CONSUMPTION		
TUSB8044A VDD (1.1V x 680mA)	0.75 W	680 mA*
TUSB8044A VDD33 (3.3V x 100mA)	0.33 W	100 mA*
LDO Loss (2.2V x 680mA)	1.50 W	---
Buck Loss (5V->3.3V, eff~90%)	0.28 W	---
HD3SS3220 x2	0.01 W	2 mA
Misc (pull-ups, LED)	0.02 W	5 mA
Total from 5V	2.91 W	582 mA
* Current on respective rails		
AVAILABLE FOR DOWNSTREAM		
3000 - 582 = 2418 mA @ 5V	12.09 W	2418 mA
PORT CURRENT LIMITS (Per Channel)		
P1 USB-C (TPS2561#1) R_ILIM=37.4k		1.5 A max
P2 USB-A (TPS2561#1) R_ILIM=37.4k		1.5 A max
P3 USB-A (TPS2561#2) R_ILIM=56.0k		1.0 A max
P4 USB-A (TPS2561#2) R_ILIM=56.0k		1.0 A max
Worst case all ports:	5.0 A	[EXCEEDS BUDGET]
Typical 2 ports:	~1.0 A	[WITHIN BUDGET]
NOTES		
- Worst case exceeds 3A upstream. PC Host overcurrent protection will limit total draw.		
- BC 1.2 CDP: 1.5A/port while data active.		
- BC 1.2 DCP: 1.5A/port, upstream absent.		

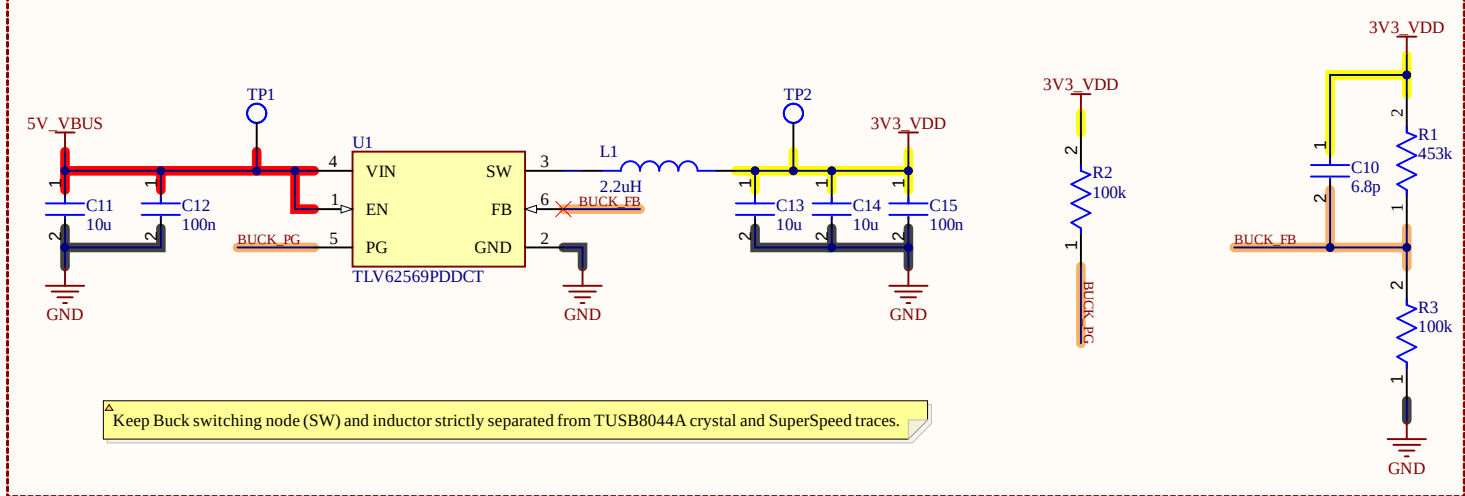
DESIGN NOTES
SUPERSPEED ROUTING OPTIMIZATION RX/TX polarity inversion (P/N swapping) has been implemented on specific downstream/upstream ports to optimize PCB routing and preserve Signal Integrity. This is fully compliant with the USB 3.2 Gen 1 specification, which mandates automatic polarity inversion detection at the receiver.

APPROVALS		DATE		PROJECT Altium™	
ENG: Alberto Marrone					
DSN: Alberto Marrone					
CHK:					
REFERENCE DOCUMENTS				PROJECT REVISION: Not in version controlDOCUMENT REVISION: Not in version controlDESIGN ITEM: Not in version control	
BOM:					
ASSY DWG:					
FAB DWG:					
PCB DWG:					
TITLE		USB 3.2 Gen 1 4-Port Hub			
SIZE	CAGE CODE	DWG NO.		REV	
B					
SCALE:		FILE NAME		SHEET	OF
		Top_Level.SchDoc		1	7

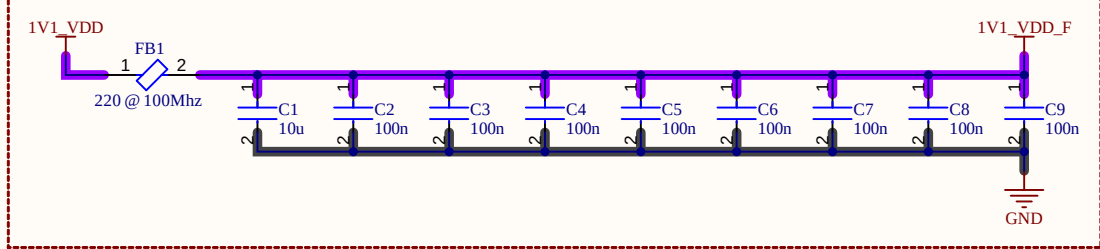
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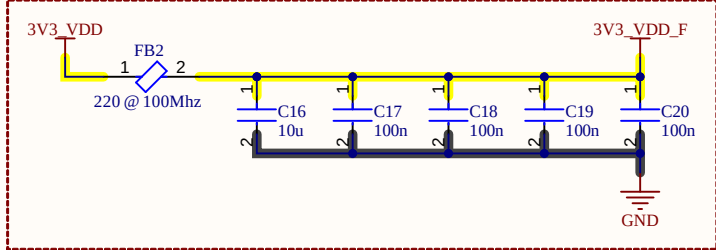
5V to 3.3V Buck Converter



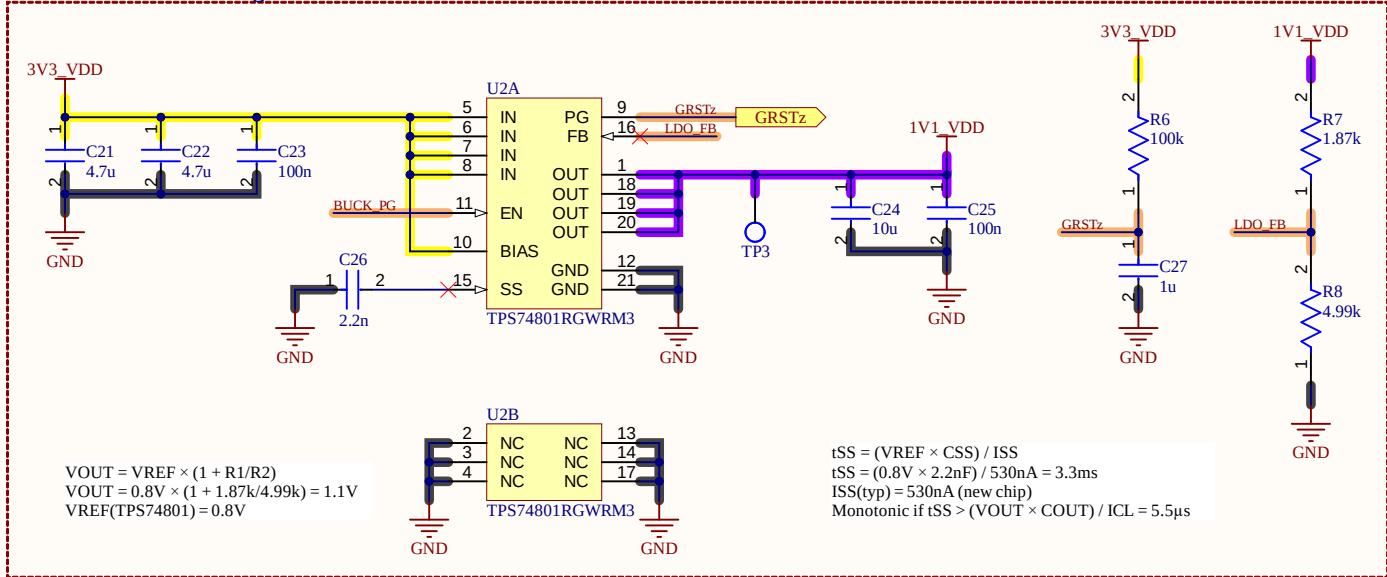
TUSB8044A Decoupling - VDD 1.1V



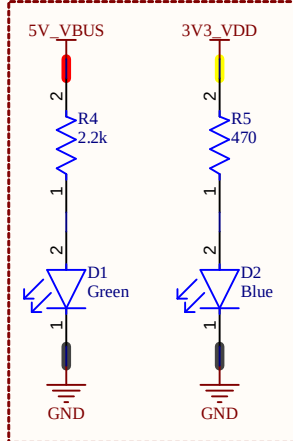
TUSB8044A Decoupling - VDD 3.3V



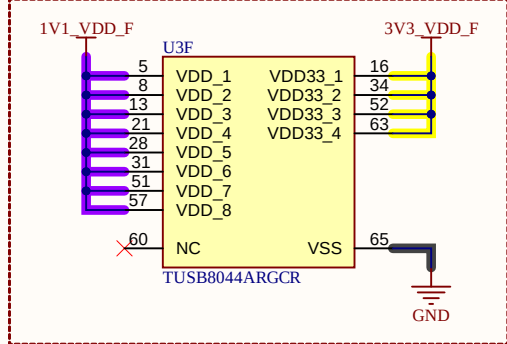
3.3V to 1.1V LDO Regulator



LED Indicators

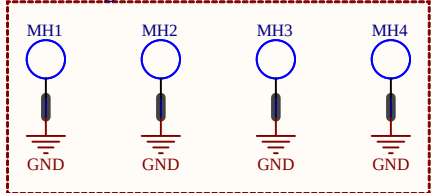


TUSB8044A Power



PG threshold $\approx 90\% \times VOUT = 0.99V$
PG HIGH $\rightarrow$ GRSTz released $\rightarrow$ TUSB8044A starts

Mounting Holes M3



Power-on sequence:
1. VBUS 5V $\rightarrow$ Buck starts (EN tied to VIN)
2. Buck PG releases $\rightarrow$ LDO EN goes HIGH ($t \approx 0.5ms$)
3. LDO soft-start complete ($t \approx 3.8ms$, $CSS=2.2nF$)
4. LDO PG releases $\rightarrow$ RC delay charges C_DELAY
5. GRSTz reaches VIH (2V) $\rightarrow$ TUSB8044A exits reset

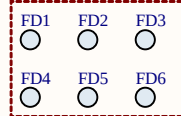
Timing analysis:
t1 = Buck startup to PG: $\sim 0.5ms$
t2 = LDO soft-start ($VREF \times CSS/ISS$): $3.3ms$
t3 = LDO PG delay (90% of 1.1V): $\sim 0.5ms$
t4 = RC delay to VIH: $11.8ms$

Total power-on time: $\sim 16ms$

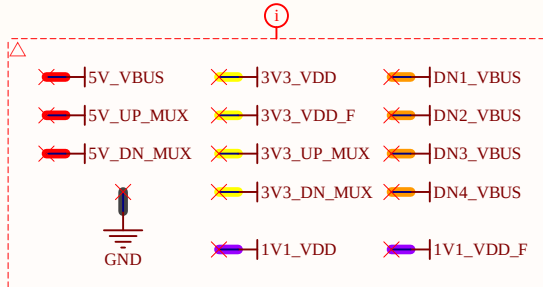
RC delay calculation (GRSTz):
 $R_{ext} = 100k\Omega$ (external pull-up on PG)
 $R_{int} = 14.5k\Omega$ min (TUSB8044A internal RPU)
 $R_{eq} = R_{ext} \parallel R_{int} = 12.66k\Omega$ (worst case)
 $C_{DELAY} = 1\mu F$
 $\tau = R_{eq} \times C = 12.66ms$
 $t(VIH=2V) = -\tau \times \ln(1 - 2/3.3) = 11.8ms$

td2 = 11.8ms > 3ms min (TUSB8044A requirement) ✓

Fiducials



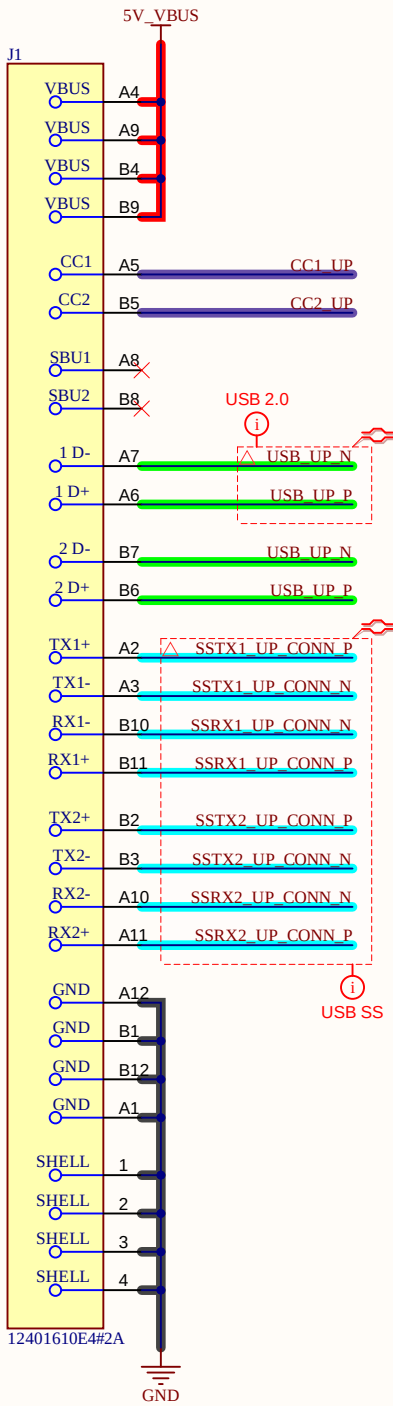
Power Nets



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REFERENCE DOCUMENTS		TITLE		
BOM:		USB 3.2 Gen 1 4-Port Hub		
ASSY DWG:	SIZE	CAGE CODE	DWG NO.	REV
FAB DWG:	B			
PCB DWG:	SCALE:	FILE NAME	Power.SchDoc	SHEET 2 OF 7

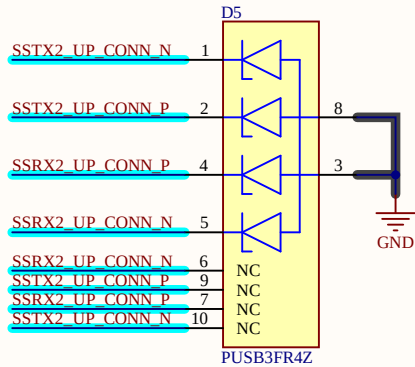
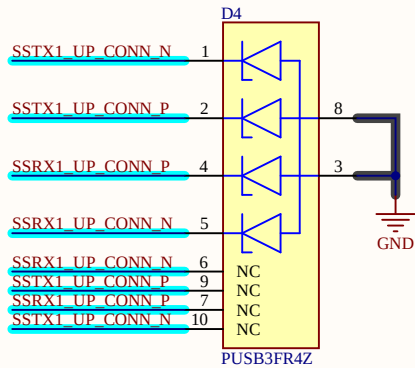
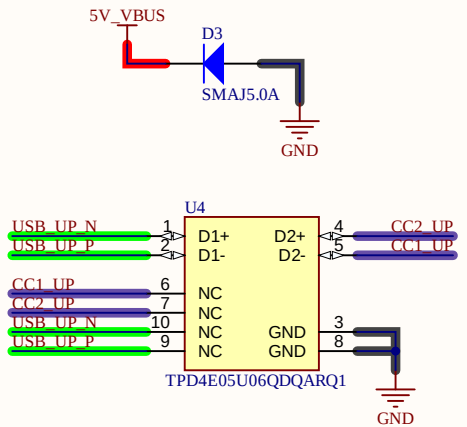
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USB-C Receptacle (Upstream)



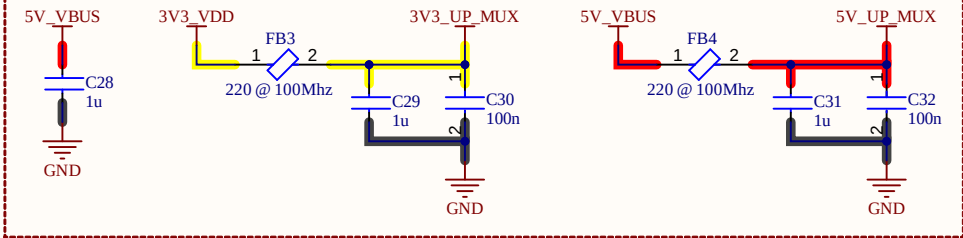
USB-C SBU1/SBU2 left floating
(no Alternate Mode)

ESD Protection

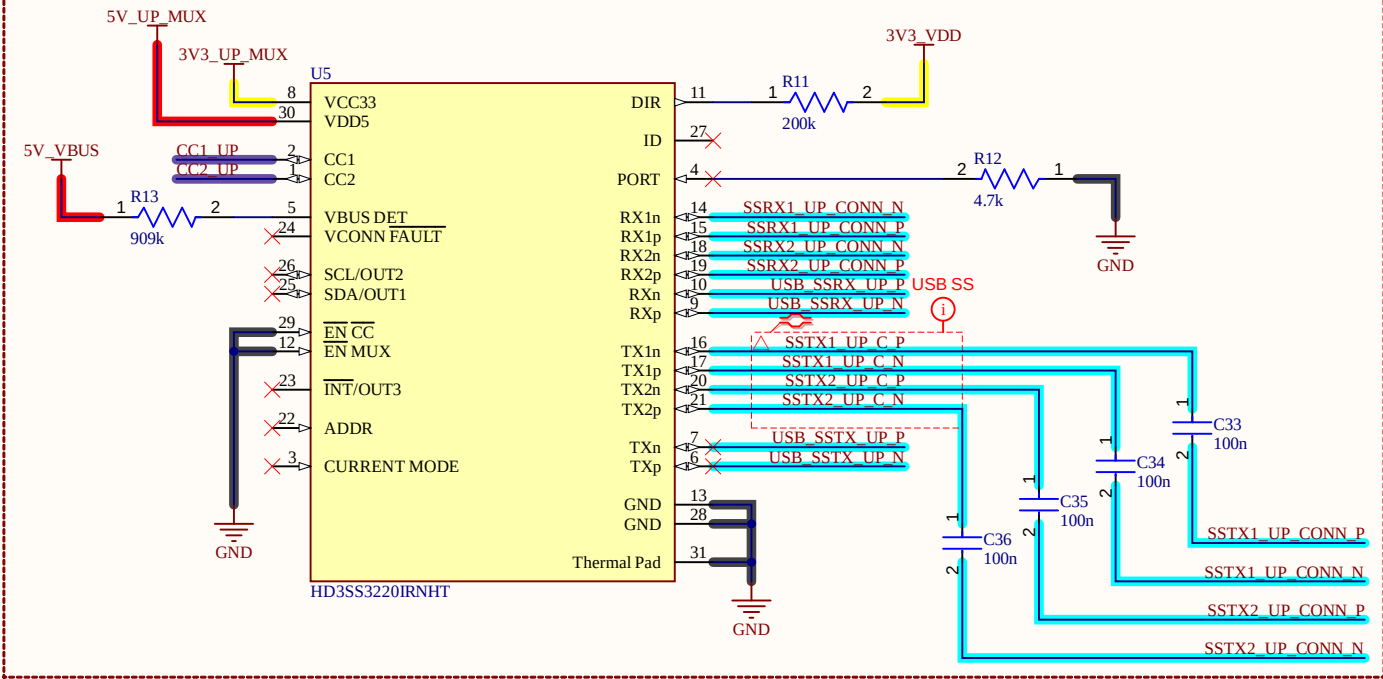


ESD Protection:
- PUSB3FR4Z: 0.29pF, ±15kV, USB3 protection
- TPD4E05U06: 0.5pF, USB2/CC protection
- SMAJ5.0A: VBUS TVS, clamps surges to ground

Power Filtering



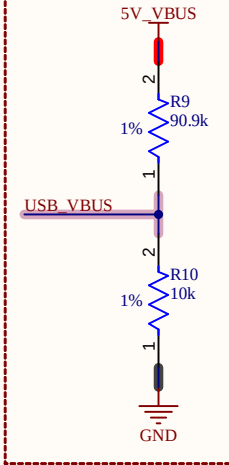
USB-C MUX & CC Controller (UFP Mode)



HD3SS3220 Configuration:
- UFP-only mode (PORT pin tied LOW via 4.7kΩ)
- GPIO mode (ADDR floating)
- ENn_CC = ENn_MUX = LOW (always enabled)
- VDD5 stable before VCC33 (datasheet requirement met)

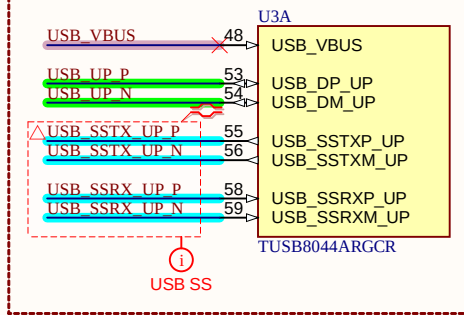
NOTE: SuperSpeed P/N Polarity Swapped for PCB routing optimization.

VBUS Detect Divider



VBUS_DET divider:
 $V_{pin} = 5V \times 95k / (909k + 95k) = 0.47V$
Within HD3SS3220 detect range

TUSB8044A Upstream Port

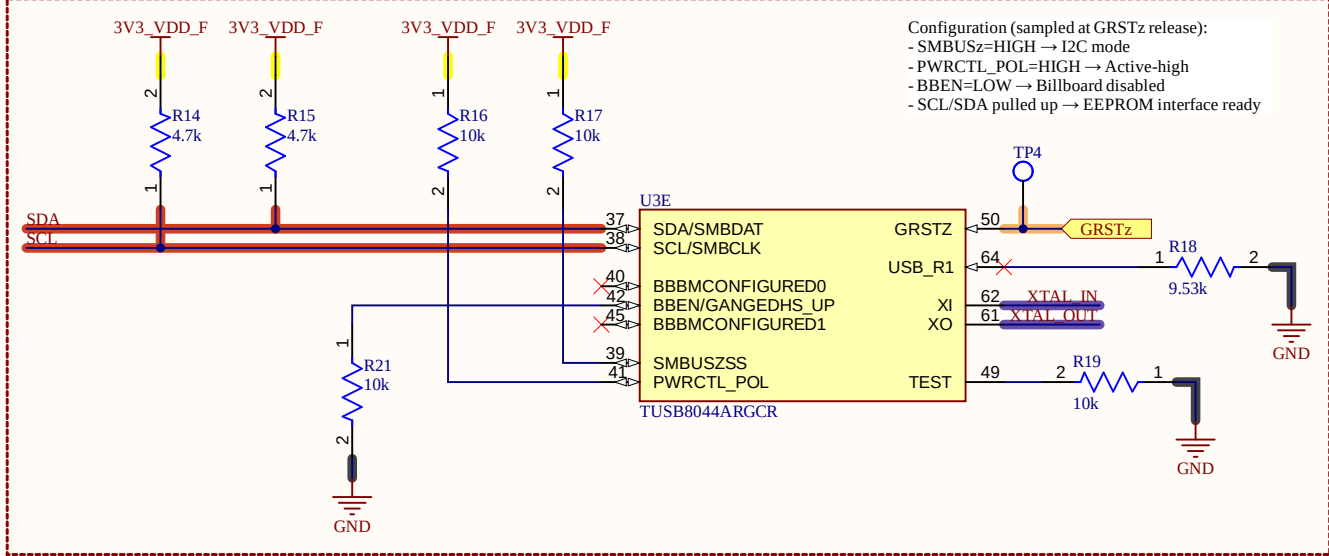


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REFERENCE DOCUMENTS		TITLE		
BOM:		USB 3.2 Gen 1 4-Port Hub		
ASSY DWG:	SIZE	CAGE CODE	DWG NO.	REV
FAB DWG:	B			
PCB DWG:	SCALE:	FILE NAME	Upstream.SchDoc	SHEET 3 OF 7

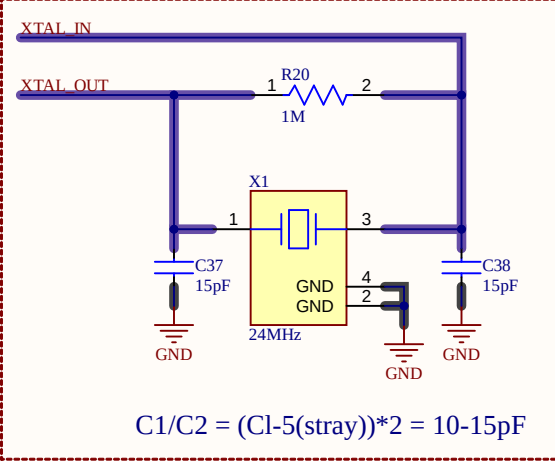
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TUSB8044A Configuration Ports



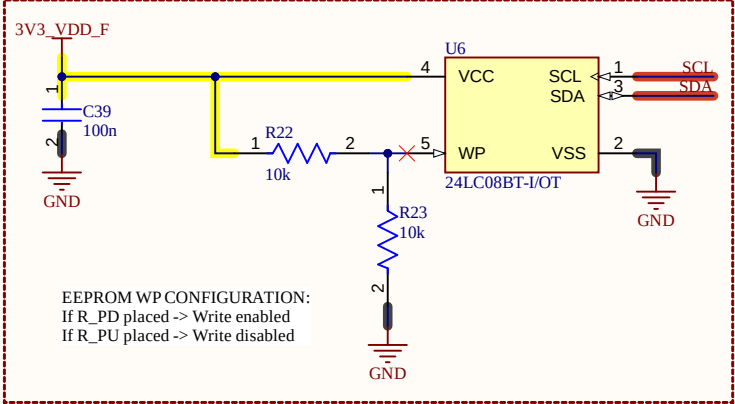
Crystal Oscillator



CRYSTAL PLACEMENT:

- Place within 5mm of TUSB8044A XI/XO pins
- Short, direct traces
- Solid GND plane underneath
- Keep switching signals away
- Guard ring with GND vias recommended

EEPROM



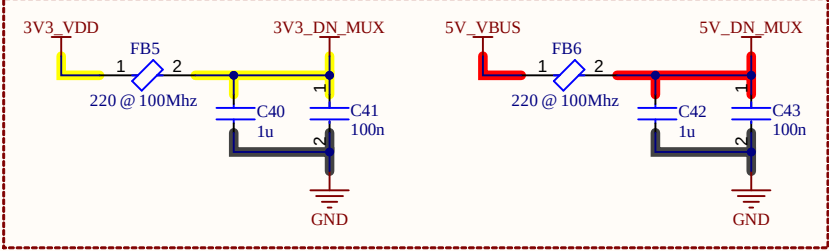
EEPROM WP CONFIGURATION:
If R_PD placed -> Write enabled
If R_PU placed -> Write disabled

WARNING: Do not populate both resistors simultaneously.

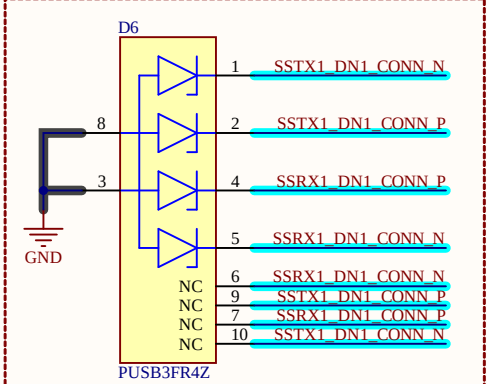
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REFERENCE DOCUMENTS			TITLE		
BOM:			USB 3.2 Gen 1 4-Port Hub		
ASSY DWG:	SIZE	CAGE CODE	DWG NO.	REV	
FAB DWG:	B				
PCB DWG:	SCALE:	FILE NAME	Hub Core.SchDoc	SHEET	4 OF 7

REVISION	DESCRIPTION	DATE	APPROVED

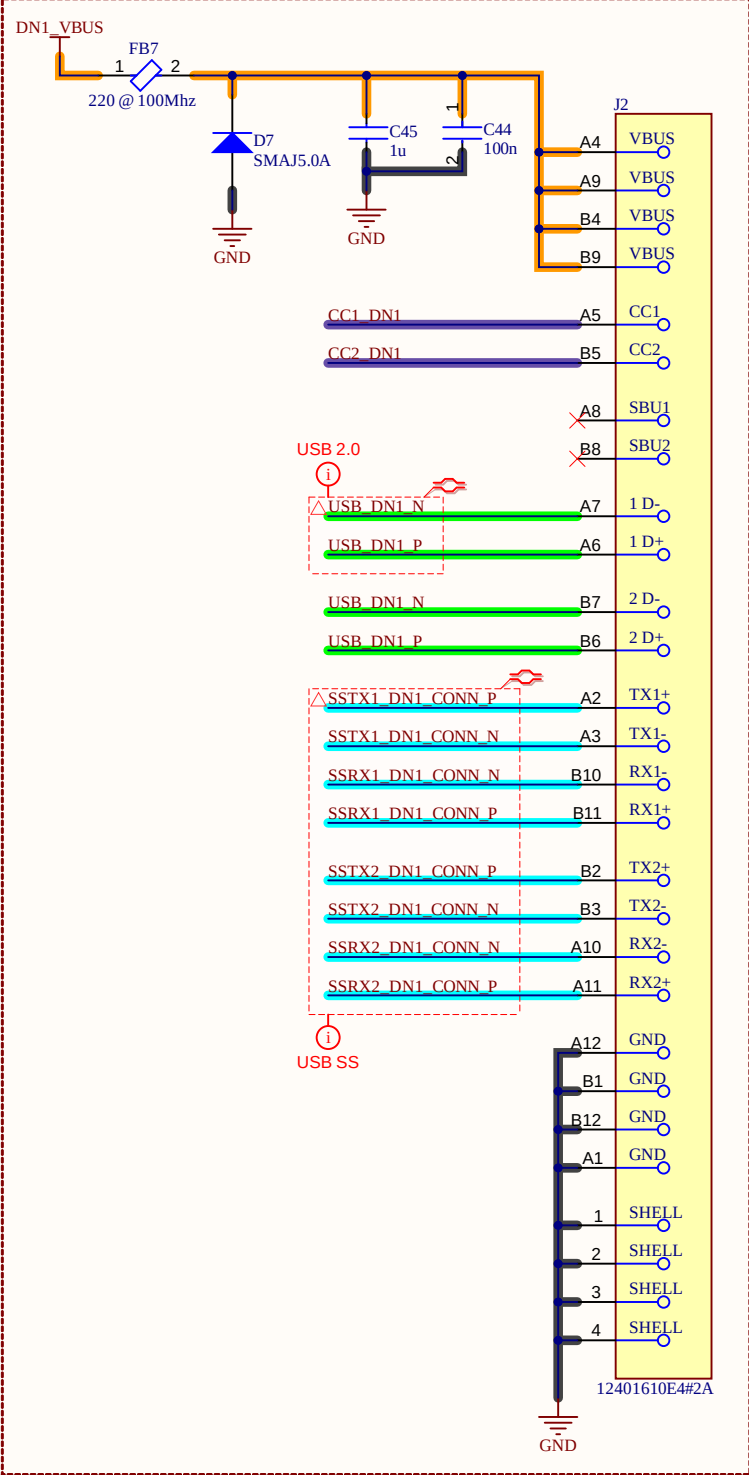
Power Filtering



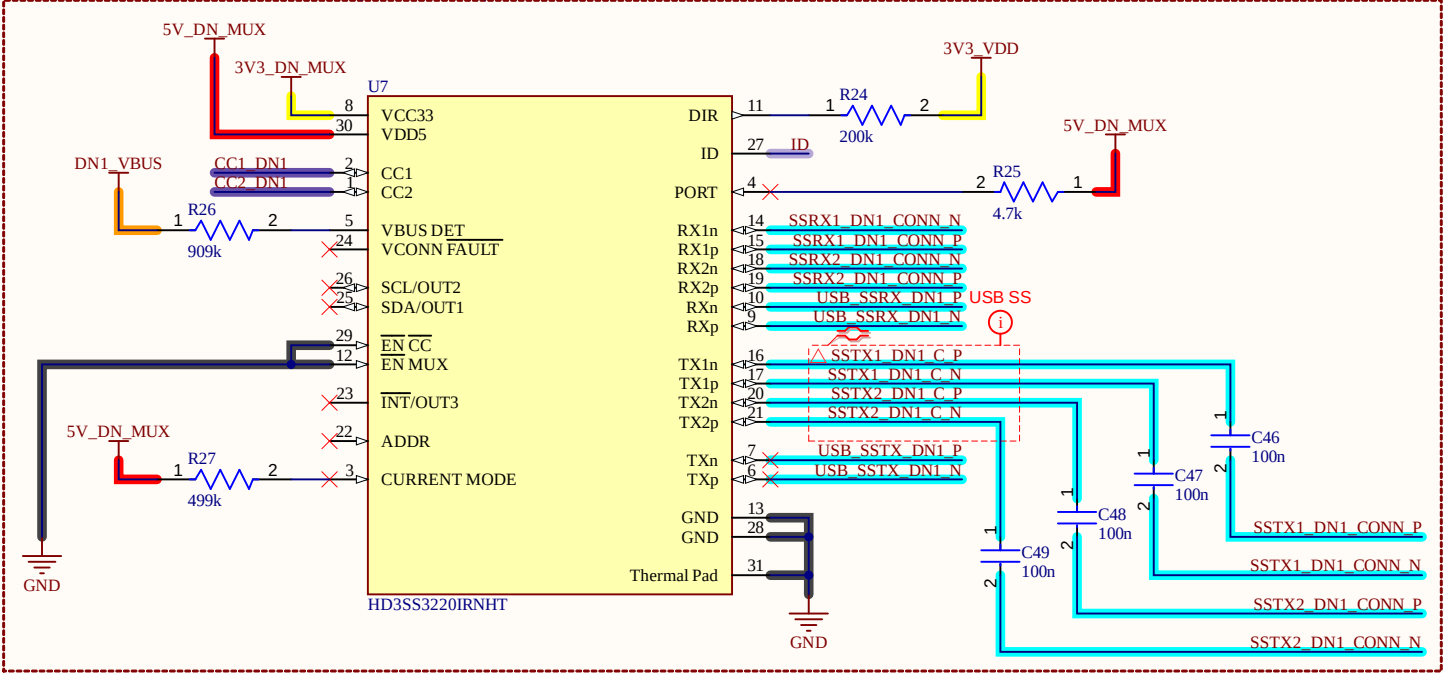
ESD Protection



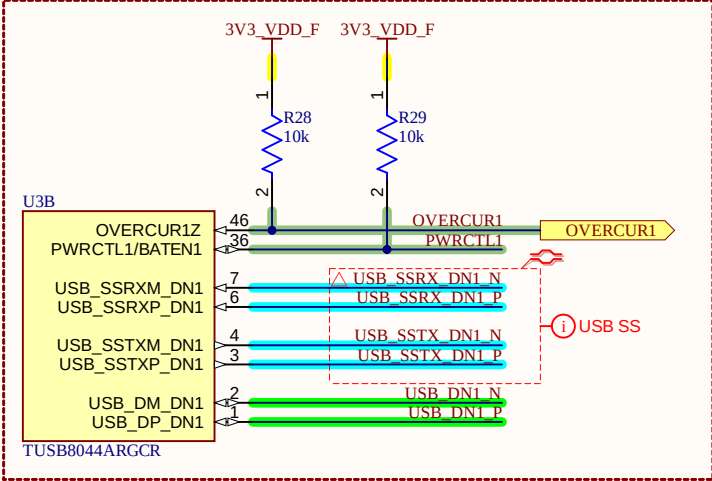
USB-C Receptacle (Downstream Port 1)



USB-C MUX & CC Controller (DFP Mode)

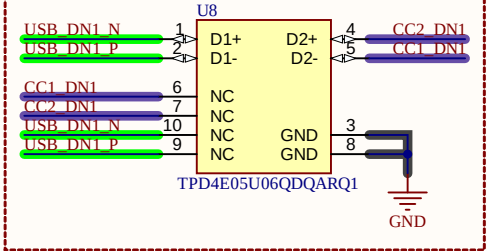
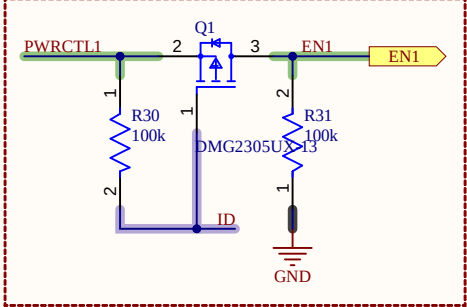


TUSB8044A Downstream Port 1



NOTE: SuperSpeed P/N Polarity Swapped for PCB routing optimization.

Cold Socket Logic (USB-C Port 1)

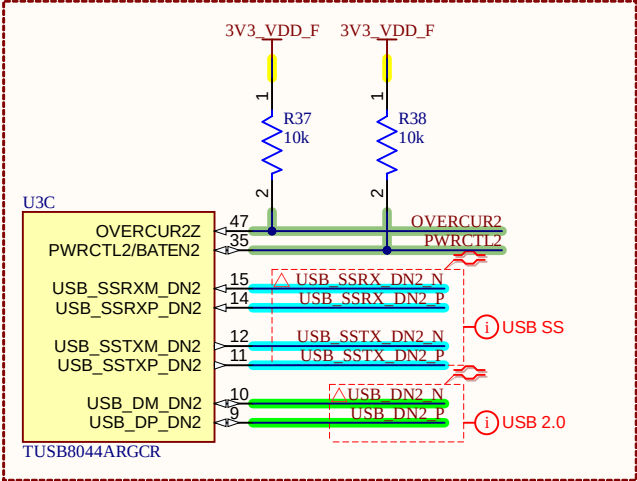


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CHK:		TITLE	USB 3.2 Gen 1 4-Port Hub
REFERENCE DOCUMENTS		SIZE	CAGE CODE
BOM:		DWG NO.	REV
ASSY DWG:		SCALE:	FILE NAME
FAB DWG:		Downstream Port 1.SchDoc	SHEET 5 OF 7
PCB DWG:			

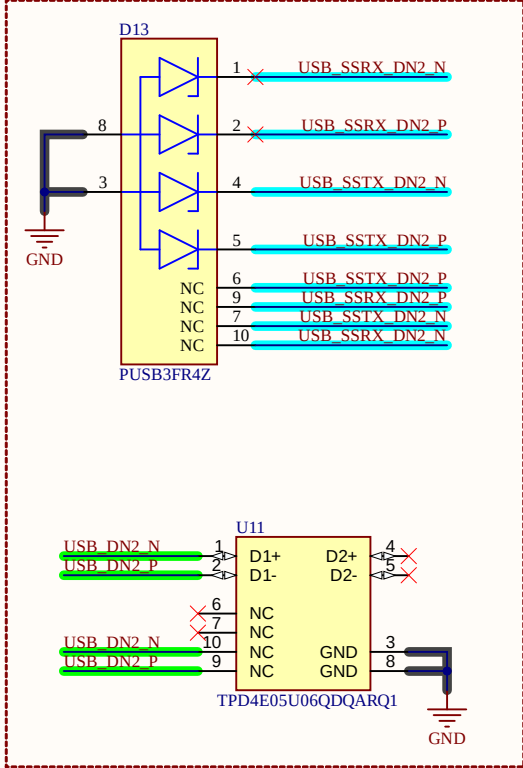
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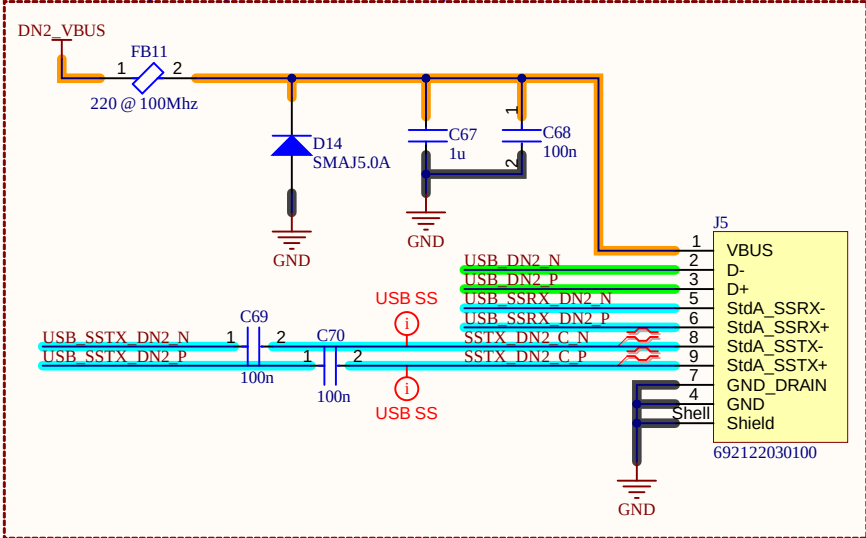
TUSB8044A Downstream Port 2



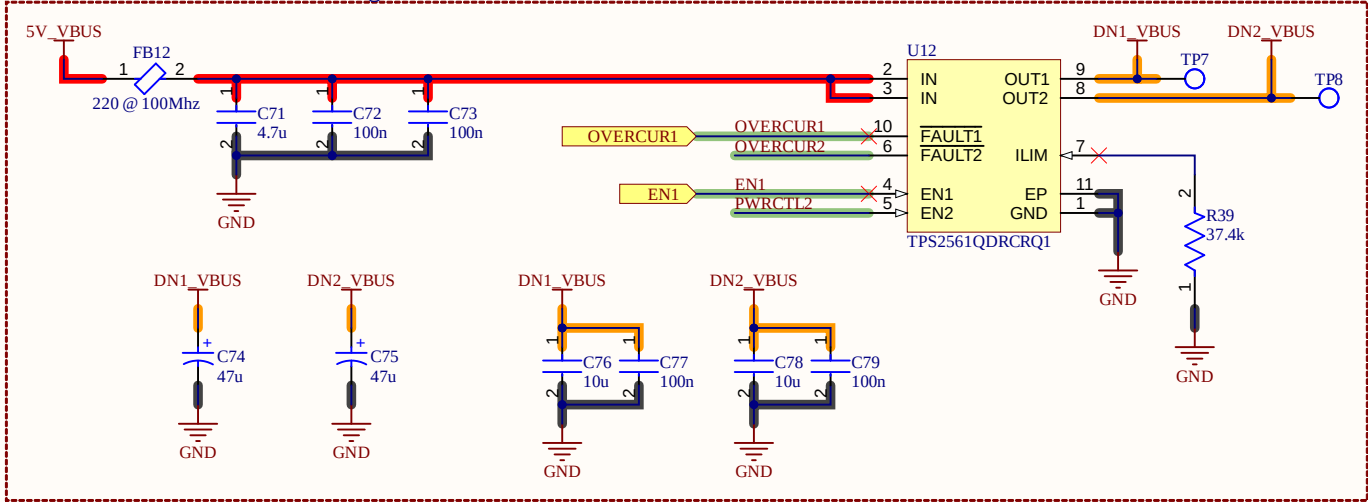
ESD Protection



USB-A Receptacle (Downstream Port 2)



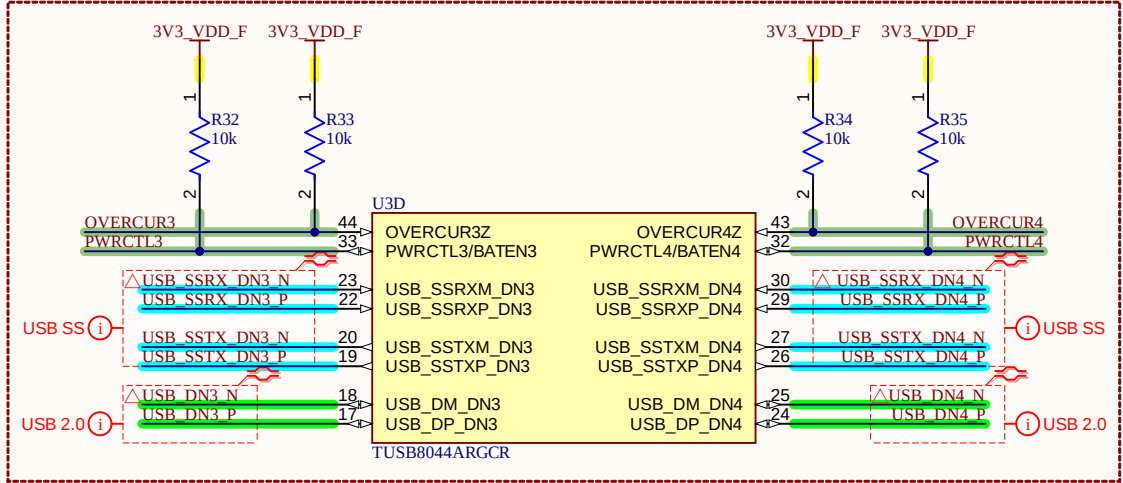
Power Switch & Current Limiting



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DSN: Alberto Marrone		CHK:	Not in version control	Not in version control	
REFERENCE DOCUMENTS			TITLE		
BOM:			USB 3.2 Gen 1 4-Port Hub		
ASSY DWG:	SIZE	CAGE CODE	DWG NO.	REV	
FAB DWG:	B				
PCB DWG:	SCALE:	FILE NAME	Downstream Port 2.SchDoc	SHEET	6 OF 7

REVISION	DESCRIPTION	DATE	APPROVED

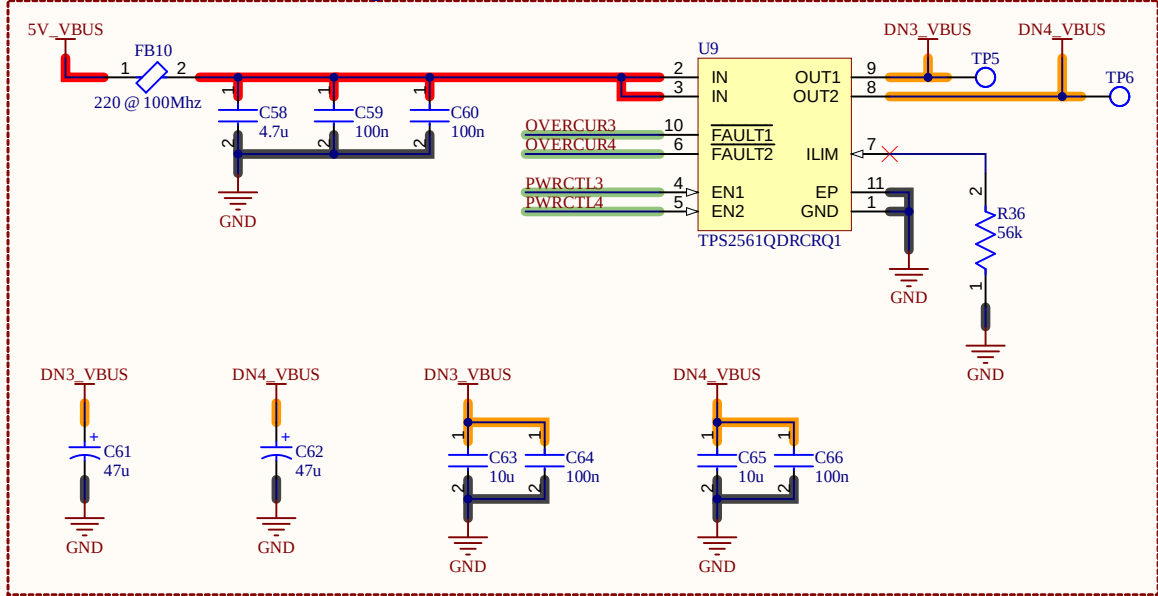
TUSB8044A Downstream Ports 3-4



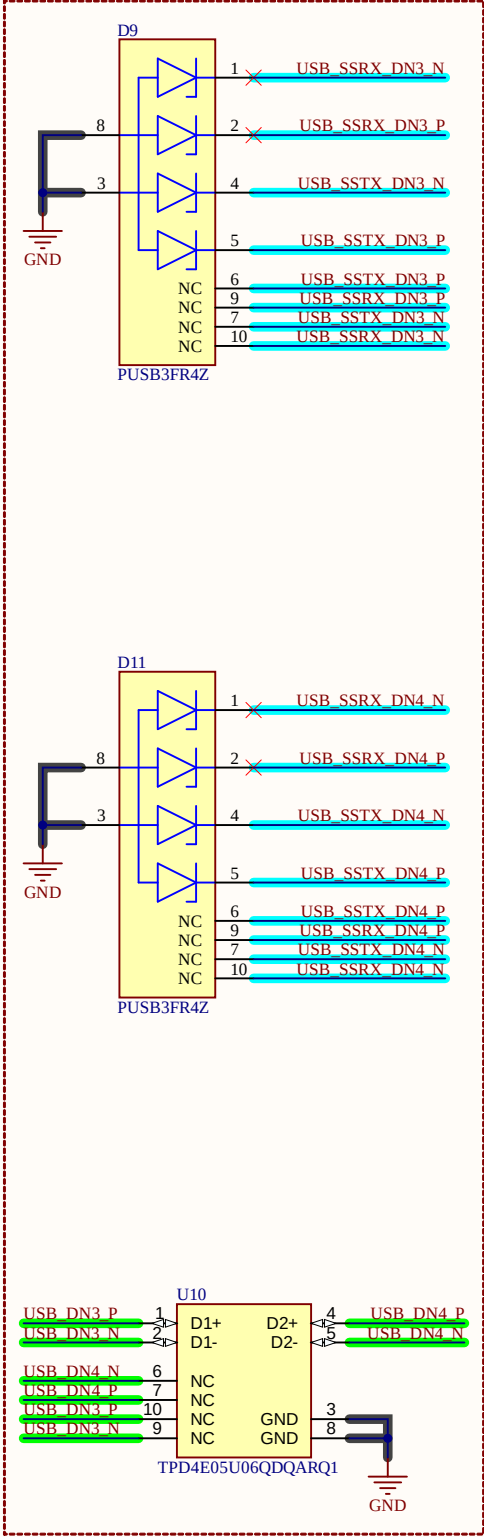
TPS2561 Current Limit Configuration:
I_OS(typ) = 56000 / R_ILIM [kΩ] (mA)
R_ILIM = 37.4kΩ → 1.5A/ch (Ports 1+2)
R_ILIM = 56kΩ → 1.0A/ch (Ports 3+4)

Per-channel limit, NOT shared between channels.
Overcurrent triggers output disconnect + FAULT LOW.
Auto-retry after fault condition clears.

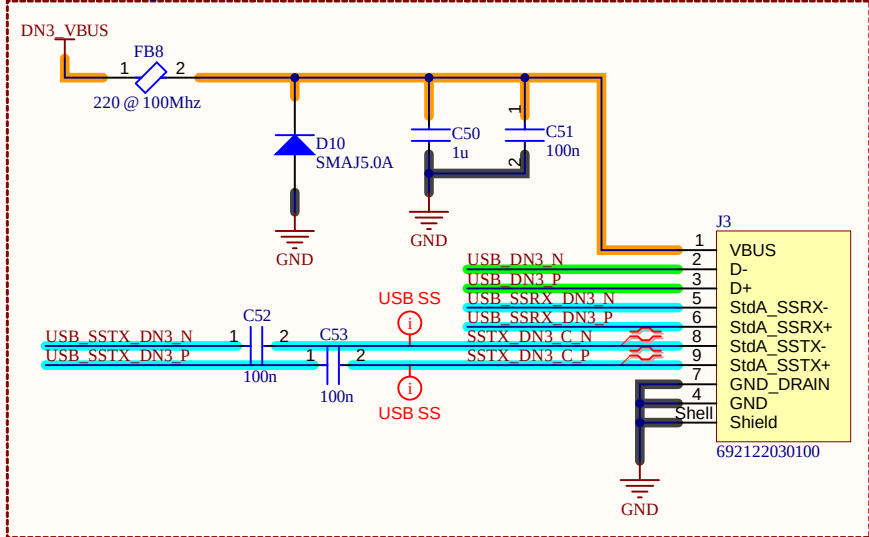
Power Switch & Current Limiting



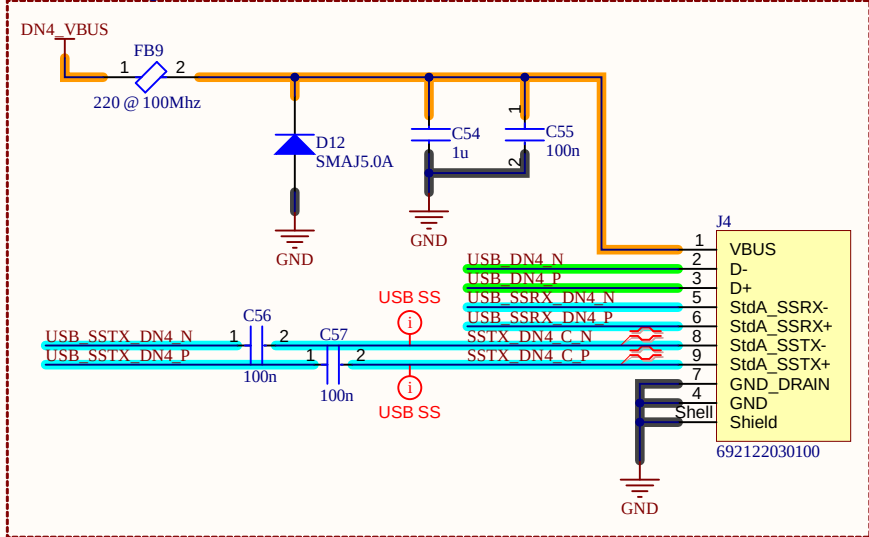
ESD Protection



USB-A Receptacle (Downstream Port 3)



USB-A Receptacle (Downstream Port 4)



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REFERENCE DOCUMENTS		TITLE		
BOM:		USB 3.2 Gen 1 4-Port Hub		
ASSY DWG:	SIZE	CAGE CODE	DWG NO.	REV
FAB DWG:	B			
PCB DWG:	SCALE:	FILE NAME	Downstream Ports 3-4.SchDoc	SHEET 7 OF 7